작성 안내 (제출 전 이 페이지 삭제)

이 초안의 구조. 결과와 무관한 부분(서론·이론, 실험 방법, 고찰의 이론적 틀, 참고문헌)은 완성된 문장으로 써 두었습니다. 실험 후 바뀌는 값과 관찰은 회색 상자로 남겼습니다. 상자 안의 숫자는 매뉴얼의 계획값이므로 **반드시 본인 측정값으로 바꿔야 합니다**. 결과에 따라 문장이 달라지는 곳은.tex 파일에 % 주석으로 선택지를 달아 두었습니다.

실험 후 채울 것

1. 2절과 표 1: 실제 부피, Grignard 적가 시간, 반응 시간, 추출 부피.
2. 3절: 조생성물과 정제 생성물 질량(두 번의 datasheet 값), 수율, TLC(반응 후 판과 컬럼 분획), IR, ^{1}H NMR 실측값.
3. 표 5의 "예상 δ"는 구조에서 추정한 대략값입니다. 문헌값으로 바꾸려면 강의 자료나 SDBS에서 확인하세요.
4. 4절 주석의 선택지 중 결과에 맞는 문장 선택.

제출 전 확인할 것 [CONFIRM]

1. 유기 파트 보고서 형식·분량·마감(평가 방식 공지 확인). proforma 형식이면 문단을 답안으로 옮겨 쓰면 됩니다.
2. 과목의 생성형 AI 사용 규정(사용 사실 표기 필요 여부).
3. 짝(Schlenk line 공유)과 반응을 각자 했는지 함께 했는지. 함께 했다면 2절 첫 문장과 수율 기준을 수정.
4. NMR 측정 주파수(400 MHz로 가정), IR 기기명, 불활성 기체 종류(질소로 가정).

컴파일: XeLaTeX. 파일 첫 줄 근처의 `\drafttrue`를 `\draftfalse`로 바꾸면 이 페이지가 빠집니다.

Synthesis of (±)-Linalool by Addition of Vinylmagnesium Bromide to 6-Methyl-5-hepten-2-one

Kim Donghwan (A0266094X)

CM3191 Chemical Synthesis Experiments, Organic Experiment 4

Thursday laboratory, fume hood B7 • Date of experiment: 15 and 29 October 2026

Instructor: Dr Hoang Truong Giang

Abstract. Racemic linalool was prepared by the addition of vinylmagnesium bromide to 6-methyl-5-hepten-2-one in anhydrous THF under a nitrogen atmosphere, using Schlenk-line techniques. After quenching with saturated ammonium chloride and extractive work-up, the crude product (0.XXX g) was purified by flash column chromatography on silica gel to give linalool as a colourless oil (0.XXX g, XX %). The structure was confirmed by the disappearance of the ketone C=O absorption and methyl ketone signal and by the appearance of the O–H absorption and the characteristic vinyl signals in the IR and ^{1}H NMR spectra.

1. Introduction

Grignard reagents, R–MgX, were discovered by Victor Grignard, who received the Nobel Prize in Chemistry in 1912.[1,2] The C–Mg bond is strongly polarised towards carbon, so the carbon atom behaves as a powerful nucleophile and base. Grignard reagents add to the electrophilic carbon of carbonyl groups: with ketones they give tertiary alcohols after protonation of the intermediate magnesium alkoxide.[3]

Linalool is a tertiary allylic alcohol and a monoterpene found in the essential oils of more than 200 plant species. (*S*)-Linalool is a major constituent of coriander oil and (*R*)-linalool of lavender oil.[2] In this experiment it is prepared by adding a vinyl group to the ketone carbonyl of 6-methyl-5-hepten-2-one (Figure 1). The carbonyl carbon is prochiral: because both the ketone and the Grignard reagent are achiral, attack on the two faces is equally likely, and the product is racemic (±)-linalool.

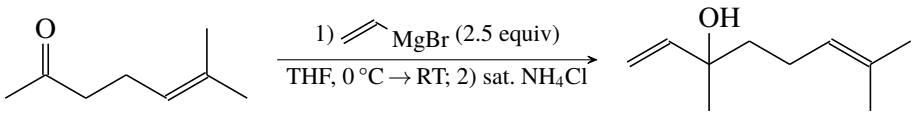


Figure 1: Synthesis of (±)-linalool (MW 154.25) from 6-methyl-5-hepten-2-one (MW 126.20) and vinylmagnesium bromide.

Because the Grignard reagent is such a strong base, it is destroyed immediately by water and other protic compounds; vinylmagnesium bromide gives ethylene on protonation.[2] It also reacts with oxygen. The reaction is therefore carried out in anhydrous THF under nitrogen, in glassware dried by heating under vacuum on a Schlenk line and filled with nitrogen by repeated evacuate–refill cycles, and the reagent is transferred by syringe through a septum.[4] An excess of the Grignard reagent (2.5 equiv) compensates for reagent lost to traces of moisture and to a competing side reaction: acting as a base, the Grignard reagent can deprotonate the methyl ketone at the α-carbon to form an enolate, which simply returns the starting ketone on work-up.[3]

The reaction is quenched with saturated aqueous ammonium chloride, a weak acid that protonates the magnesium alkoxide and destroys the excess Grignard reagent. A strong acid is avoided because tertiary allylic alcohols such as linalool are prone to acid-catalysed dehydration and rearrangement.[3] The crude product is

purified by flash column chromatography.[5] Since neither the starting material nor the product absorbs strongly in the UV, TLC plates are visualised with potassium permanganate, which reacts with the C=C bonds present in both compounds.[2]

The aims of this experiment were (i) to carry out a Grignard addition under an inert atmosphere using Schlenk-line and syringe techniques, (ii) to purify the product by flash column chromatography guided by TLC, and (iii) to confirm the structure of linalool by IR and ^{1}H NMR spectroscopy and to determine the yield.

2. Experimental

2.1 General

6-Methyl-5-hepten-2-one, vinylmagnesium bromide (1.0 M in THF, supplied under nitrogen), anhydrous THF, saturated aqueous NH_4Cl, ethyl acetate, hexane, brine, anhydrous Na_2SO_4 and silica gel were used as supplied. Reactions were carried out under [nitrogen] on a Schlenk line. TLC was performed with hexane–ethyl acetate (9:1) and visualised with $KMnO_4$ stain. IR spectra were recorded on [instrument] with an ATR accessory. ^{1}H NMR spectra were recorded at [400] MHz in $CDCl_3$; chemical shifts are reported relative to residual $CHCl_3$ (δ 7.26).[6]

2.2 Grignard reaction (session 1)

A 50 mL two-necked round-bottom flask containing a stirrer bar was fitted with a rubber septum on the side neck and connected to the Schlenk line through a stopcock adapter on the central neck. The flask was evacuated, heated with a hot-air gun, allowed to cool and refilled with nitrogen; this cycle was performed twice, and the flask was allowed to cool to room temperature. 6-Methyl-5-hepten-2-one (0.44 mL, [0.376] g, [2.98] mmol) and anhydrous THF (5 mL) were added, and the solution was cooled in an ice–water bath for 5–10 min. Vinylmagnesium bromide (1.0 M in THF, [7.5] mL, [7.5] mmol, [2.5] equiv) was withdrawn with a nitrogen-purged syringe and added dropwise over [X] min. [OBS: colour, exotherm] The ice bath was removed and the mixture was stirred at room temperature for 90 min.

2.3 Work-up

The flask was disconnected from the Schlenk line, closed with septa and cooled in an ice–water bath, and saturated aqueous NH_4Cl (10 mL) was added dropwise. [OBS: gas, precipitate] Water (10 mL) was added, the mixture was transferred to a separatory funnel with ethyl acetate (20 mL), and the layers were separated. The aqueous layer was extracted with two further portions of ethyl acetate (20 mL each). The combined organic layers (about 60 mL) were washed with brine (15 mL) and dried over anhydrous Na_2SO_4 for 15 min. TLC of the dried solution against the starting ketone showed [TLC result]. The solution was filtered into a pre-weighed 250 mL round-bottom flask and concentrated on a rotary evaporator to give the crude product as a [pale yellow oil] ([0.XXX] g).

2.4 Flash column chromatography (session 2)

A column of silica gel ([12–15] cm) was packed as a slurry in hexane–ethyl acetate (9:1) between layers of sand. The crude product was loaded in a minimal volume of eluent (with rinses of about 1 mL and 2–3 mL), and the column was eluted with hexane–ethyl acetate (9:1, about 200 mL) under air pressure. Fractions were collected in test tubes and analysed by TLC ($KMnO_4$ stain). Fractions [X–Y] containing the product were

combined and concentrated on a rotary evaporator, and the residue was dried under vacuum for 20 min to give (±)-linalool as a colourless oil (0.XXX g, X.XX mmol, XX %).

Characterisation data. IR (ATR) $\tilde{\nu}_{\max}$/1/cm: values. ^{1}H NMR (400 MHz, $CDCl_3$) δ: values.

3. Results

Table 1: Quantities of reagents (planned values from the laboratory manual[2] and actual values used).

Reagent	M / g mol^{-1}	Density / conc.	Planned	Actual	n / mmol	Equiv.
6-Methyl-5-hepten-2-one	126.20	0.855 g mL^{-1}	0.44 mL	☐	2.98	1.00
Vinylmagnesium bromide	131.25	1.0 M in THF	7.5 mL	☐	7.5	2.5
THF (anhydrous)	72.11	—	5 mL	☐	solvent	—
(±)-Linalool	154.25	—	theor. 0.460 g			

Table 2: Masses and yields.

	Crude (session 1)	Purified (session 2)
Mass of empty flask / g	☐	☐
Mass of flask + product / g	☐	☐
Mass of product / g	☐	☐
Yield (theoretical 0.460 g)	XX % (crude)	XX %

Table 3: TLC data (silica, hexane–EtOAc 9:1, $KMnO_4$ stain; solvent front X.X cm).

Sample	Distance / cm	R_f	Comment
6-Methyl-5-hepten-2-one	☐	☐	
Reaction mixture (dried organic layer)	☐	☐	OBS: new spot? SM left?
Column fractions containing product	☐	☐	OBS: fractions X–Y

Table 4: Principal IR absorptions: disappearance of the ketone and appearance of the alcohol.

Vibration	Typical range[7] / 1/cm	Starting ketone	Product (obs.)
O–H stretch (H-bonded)	3500–3200 (broad)	absent	☐
C=O stretch (aliphatic ketone)	ca. 1715	present	absent?
C=C stretch	ca. 1680–1620	weak	☐
=C–H out-of-plane bend, $-CH{=}CH_2$	ca. 990 and 910	absent	☐

Table 5: ^{1}H NMR data of the product ($CDCl_3$). Expected values are approximate positions from the structure; the vinyl pattern is based on closely related linalool derivatives.[8]

Proton(s)	Expected δ	Mult.	Obs. δ / $\int$	Comment
$-CH{=}CH_2$ (internal H)	ca. 5.9	dd, $J \approx 17$, 11 Hz	☐	new; not in the ketone
$=CH_2$ (*trans* to CH)	ca. 5.2	dd, $J \approx 17$, 1 Hz	☐	new
$=CH_2$ (*cis* to CH)	ca. 5.1	dd, $J \approx 11$, 1 Hz	☐	new
$-CH{=}C(CH_3)_2$	ca. 5.1	br t	☐	also present in the ketone
allylic CH_2	ca. 2.0	m	☐	
CH_2–C(OH)	ca. 1.5	m	☐	
$=C(CH_3)_2$	ca. 1.7, 1.6	$2 \times$ s	☐	also present in the ketone
$C(OH)CH_3$	ca. 1.3	s	☐	replaces the $COCH_3$ singlet near 2.1
OH	variable	br s	☐	exchangeable

Table 6: Observations.

Stage	Observation
Drying the flask on the Schlenk line	OBS: condensation seen and removed?
Addition of vinylmagnesium bromide	OBS: colour, warming, rate
Quench with NH_4Cl	OBS: gas, solids
Extraction	OBS: layers, emulsions
Crude product	OBS: appearance, smell
Column	OBS: band behaviour, number of fractions
Purified product	OBS: appearance, smell

4. Discussion

4.1 Inert-atmosphere technique

Drying the flask by heating under vacuum removes the thin film of water adsorbed on the glass surface. Repeating the evacuate–refill cycle reduces the amount of residual air further with each cycle.[4] Purging the syringe with nitrogen and drawing a small nitrogen cushion into it before withdrawing the needle kept the reagent from contacting air during transfer.[2]

Comment on anything that may have admitted moisture or air (e.g. a leaking septum, a slow nitrogen flow) and on how the reaction looked.

4.2 Evidence for the formation of linalool

The key spectroscopic changes are those of the reacting carbonyl group, since the trisubstituted alkene is present in both the starting material and the product. In the IR spectrum, the strong ketone C=O band near 1715 1/cm was absent, and a broad O–H band appeared at X 1/cm, as expected for conversion of the ketone into an alcohol.[7] In the ^{1}H NMR spectrum, the methyl ketone singlet (near δ 2.1) was replaced by a methyl singlet at δ 1.3X on the carbinol carbon. Three new one-proton signals at δ 5.9X, 5.2X and 5.0X show the typical pattern of a monosubstituted alkene. The internal vinyl proton couples with both terminal protons, with a large *trans* coupling (about 17 Hz) and a smaller *cis* coupling (about 11 Hz), and therefore appears as a doublet of doublets. Each terminal proton appears as a doublet of doublets with one of these couplings and a very small geminal coupling. This is consistent with the vinyl carbinol fragment of linalool derivatives.[8]

4.3 Chromatography

On silica gel, the alcohol is expected to be more polar than the ketone because it can act as a hydrogen-bond donor, so it should elute later and have a lower R_f. The observed R_f values were X.XX (ketone) and X.XX (product). Comment on separation. Because neither compound is UV-active, every fraction had to be checked by $KMnO_4$ staining, which detects the C=C bonds.

4.4 Yield

The crude and purified yields were XX % and XX %, respectively.

Discuss losses: Grignard reagent destroyed by moisture; enolisation of the ketone (starting material recovered); incomplete extraction; mixed fractions discarded on the column; evaporation of the relatively volatile product during rotary evaporation and vacuum drying.

Because the product is formed as a racemate, its optical rotation would be zero, in contrast to the enantiomerically enriched natural samples from coriander or lavender.

5. Conclusion

(±)-Linalool was synthesised by the Grignard addition of vinylmagnesium bromide to 6-methyl-5-hepten-2-one under a nitrogen atmosphere and purified by flash chromatography in XX % yield. IR and ^{1}H NMR spectroscopy confirmed the conversion of the methyl ketone into the tertiary allylic alcohol, in particular through the loss of the C=O band and the methyl ketone singlet and the appearance of the O–H band and the vinyl signals. The experiment demonstrates how Schlenk-line and syringe techniques allow air- and moisture-sensitive organometallic reagents to be used for carbon–carbon bond formation.

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